ATANT: An Evaluation Framework for AI Continuity

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Abstract

We present ATANT (Automated Test for Acceptance of Narrative Truth), an open evaluation framework for measuring *continuity* in AI systems the ability to persist, update, disambiguate, and reconstruct meaningful context across time. While the AI industry has produced memory components (RAG pipelines, vector databases, long context windows, profile layers), no published framework formally defines or measures whether these components produce genuine continuity. We define continuity as a system property with 7 required properties, introduce a 10-checkpoint evaluation methodology that operates without an LLM in the evaluation loop, and present a narrative test corpus of 250 stories comprising 1,835 verification questions across 6 life domains. We evaluate a reference implementation across 5 test suite iterations, progressing from 58% (legacy architecture) to 100% in isolated mode (250 stories) and 100% in 50-story cumulative mode, with 96% at 250-story cumulative scale. The cumulative result is the primary measure: when 250 distinct life narratives coexist in the same database, the system must retrieve the correct fact for the correct context without cross-contamination. ATANT is system-agnostic, model-independent, and designed as a sequenced methodology for building and validating continuity systems. The framework, story corpus, and evaluation protocol are publicly available at https://github.com/Kenotic-Labs/ATANT.

1 Introduction

Most AI systems today are session-based. A user provides input, the system responds, and the moment ends. Whatever persists is typically prompt context, conversation history, or retrieved notes. This is adequate for single-turn tasks but insufficient for any system that claims to maintain a meaningful relationship with a user over time.

Human life is not made of isolated prompts. It is made of unfinished situations, changing states, recurring concerns, relationships, timing, logistics, moods, plans, identities, and commitments. AI systems that serve humans across time need more than intelligence-in-the-moment. They need presence-over-time.

The industry has produced partial solutions: long context windows keep recent material alive temporarily; RAG pipelines retrieve semantically similar text from storage; profile layers hold static preferences; vector databases store embeddings for similarity search. None of these, individually or combined, produce what we term continuity the system property that determines what persists, in what form, what has changed, what still matters, and how to reconstruct it.

Despite growing recognition of this gap (Logan, 2026; Natangelo, 2025; Packer et al., 2023; Chhikara et al., 2025), the field lacks three things:

1. A formal definition of what continuity means as a system property.
2. A set of testable requirements that any continuity system must satisfy.

3. A benchmark that measures continuity rather than retrieval accuracy alone.

ATANT addresses all three. Our contributions are:

- A formal definition of continuity as an architectural layer, distinct from memory, retrieval, and context, with 7 required properties.
- A model-independent evaluation methodology with 10 checkpoints that tests the write path and read path of a continuity system without an LLM in the evaluation loop.
- A narrative test corpus of 250 stories (1,835 questions) across 6 life domains, designed for progressive evaluation from isolated correctness to disambiguation at scale.
- 4 compliance levels (Core, Stress, Cumulative, Scale) that provide a sequenced roadmap for building and validating continuity systems.
- Reference implementation results across 5 test suite iterations, including honest reporting of failures, limitations, and the active frontier at 96% cumulative-scale accuracy.

2 Related Work

2.1 Memory Systems for AI

Several systems have been proposed to give AI persistent memory. MemGPT (Packer et al., 2023) introduces an operating system metaphor with tiered memory (core memory always in context, archival memory stored separately). Mem0 (Chhikara et al., 2025) provides a production-oriented memory layer that extracts facts, stores them, and manages updates across user, session, and agent scopes. A-MEM (Xu et al., 2025) proposes agentic memory with self-organizing capabilities for LLM agents.

These systems address components of persistence but do not formally define or test for continuity as a system property. A system can store and retrieve facts without satisfying disambiguation, reconstruction, or temporal ordering.

2.2 Architectural Frameworks

The Continuum Memory Architecture (CMA) (Logan, 2026) defines 6 behavioral properties for long-horizon LLM agents: persistence, selective retention, retrieval-driven mutation, associative routing, temporal continuity, and consolidation. This is the closest prior work to our formalization. However, CMA focuses on memory mechanisms rather than the higher-level logic of what persists and reconstructs, and does not provide an evaluation corpus.

The Narrative Continuity Test (Natangelo, 2025) proposes a conceptual framework for evaluating identity persistence across 5 dimensions. It provides theoretical grounding but no implementation or test corpus.

2.3 Evaluation Benchmarks

Existing benchmarks evaluate specific capabilities: MemoryBench (Ai et al., 2025) tests memory and continual learning; BEAM (Tavakoli et al., 2025) benchmarks long-term memory beyond a million tokens. These evaluate memory retrieval in isolation. ATANT differs in three ways: (1) it tests the full write-path-to-read-path pipeline, not retrieval alone; (2) it uses naturalistic multi-turn narratives, not synthetic fact pairs; (3) its cumulative mode tests disambiguation under memory load a property no existing benchmark measures.

2.4 Positioning

ATANT is not a replacement for memory systems or retrieval benchmarks. It is a framework for evaluating whether the combination of components in a system produces continuity a higher-order property that emerges from correct persistence, update handling, temporal ordering, disambiguation, and reconstruction working together.

3 Defining Continuity

Continuity is the system property that enables an AI to carry forward what still matters from prior interactions, update it when reality changes, and reconstruct useful context later in the right form, at the right time, for the right situation.

3.1 Continuity vs. Memory

Memory stores the past. Continuity keeps the right parts of the past alive in the present. A database can store (user, partner_name, Mia). A continuity system can answer "Tell me about my relationship" with a reconstruction that includes Mia, where she lives, how the user feels about her, what changed last week, and what remains unresolved.

3.2 Continuity vs. Retrieval

Retrieval says: *here are some related past things.* Continuity says: *here is the current living state of the situation, including what changed, what still matters, and what should happen next.* The difference is between searching and understanding.

3.3 The 7 Required Properties

We define 7 properties that any system claiming continuity must satisfy. These properties were derived empirically by building a continuity system, running it against hundreds of real-world narratives, and identifying what breaks when each property is absent.

Table 1: The 7 Required Properties of Continuity

#	Property	Testable Requirement
1	**Persistence**	After ingesting facts, terminate and restart. All facts must be retrievable with identical accuracy.
2	**Update Handling**	Ingest a fact, then an update. System must return current state and distinguish from previous.
3	**Temporal Ordering**	Ingest facts with temporal references. System must return resolved dates, sequencing, and status.
4	**Disambiguation**	Ingest overlapping narratives. System must return correct facts without cross-contamination.
5	**Reconstruction**	After multi-turn ingestion, system must retrieve connected facts sufficient to reconstruct a situation.
6	**Model Independence**	Ingest with one model (or none). Retrieve with another. Accuracy must not degrade.

7	**Operational Usefulness**	System must function across 2+ distinct domains without architectural modification.

4 ATANT: Framework Design

4.1 Design Principles

Model Agnosticism. ATANT evaluates the continuity layer, not the intelligence layer above it. No model language, vision, or otherwise is included in the evaluation loop. The continuity layer must be correct independent of whatever intelligence layer sits on top of it.

Narrative Realism. Test inputs are naturalistic multi-turn conversations. People do not speak in database format. A single utterance may contain identity, event, time, emotional state, entity, intent, and logistics simultaneously.

Write Path + Read Path Verification. Both directions are tested: did the system correctly store the facts (write path), and did it correctly retrieve and reconstruct the answer (read path)?

Determinism. Same input, same output, every time. No sampling variance.

Progressive Difficulty (The Sequence). Testing proceeds in phases: isolated → stress → cumulative → scale. Each phase tests a harder property. The sequence tells a team where their system stands and what to fix next.

4.2 The 10-Checkpoint System

ATANT defines 10 checkpoints grouped into write-path verification (CP1–CP4), read-path verification (CP5–CP8), and cross-cutting concerns (CP9–CP10).

Table 2: ATANT Checkpoint System

CP	Path	Name	Pass Criteria
1	Write	Input Classification	Classification matches expected type
2	Write	Fact Extraction & Storage	All expected keywords in storage
3	Write	Predictive Indexing	≥1 predicted query per stored fact
4	Write	Type Tagging	Type tags match expected categories
5	Read	Query Classification	Query type matches expected
6	Read	Structural Matching	Correct fact in top-k candidates
7	Read	Convergence	Multi-fact candidate set returned
8	Read	**Final Answer**	**All expected keywords in answer**
9	Cross	Temporal Reasoning	Temporal type and direction correct
10	Cross	Contextual Adaptation	Emotion detection and direction correct

CP8 (Final Answer) is the definitive checkpoint. All others are diagnostic they identify where failures occur when CP8 fails.

4.3 Compliance Levels

ATANT defines 4 compliance levels with 3 scoring tiers (Gold: 100%, Silver: 95–99%, Bronze: 90–94%):

Table 3: ATANT Compliance Levels

Level	Requirement	What It Proves
ATANT-Core	50 stories, isolated	Basic continuity across 6 life domains
ATANT-Stress	250 stories, isolated	Continuity generalizes to novel patterns
ATANT-Cumulative	50 stories, cumulative	Disambiguation when narratives coexist
ATANT-Scale	250 stories, cumulative	Disambiguation at scale

5 Narrative Test Corpus

5.1 Design Philosophy

ATANT stories are narrative simulations realistic multi-turn conversations spanning simulated hours, days, or weeks. They test continuity for human life: personal, private, emotional, and ongoing. The domains were chosen because continuity is fundamentally about carrying a person's life forward, not task management.

Stories systematically include adversarial patterns: multi-fact utterances, shared-subject constructions ("My brother and I"), pronoun chains, temporal updates ("Actually, it moved to Thursday"), general knowledge traps ("What's the capital of France?"), emotional overlays, negation, and ambiguous predicates.

5.2 Corpus Statistics

Table 4: ATANT Corpus Statistics

Phase	Stories	Questions	Purpose
Phase 1 (Core)	50	304	6 life domains
Phase 2 Round 2	50	367	Generalization
Phase 2 Round 3	50	386	Novel patterns
Phase 2 Round 4	50	380	Edge cases
Phase 2 Round 5	50	398	Adversarial
Total	**250**	**1,835**	

5.3 Life Domain Coverage

The corpus covers 6 domains: Career (interviews, promotions, layoffs), Relationships (partners, family, friendships), Health (medical, fitness, recovery), Learning (courses, certifications, study), Daily Life (routines, errands, hobbies), and Life Events (moves, births, deaths, marriages, milestones).

5.4 Story Format

Each story contains: metadata (ID, category, simulated duration), conversation batches with simulated timestamps, expected memory stores per batch, and verification questions with expected keywords. A

question passes if all expected keywords appear in the system's retrieved answer (case-insensitive, substring-permissive).

6 Experiments and Results

We evaluate the NURA Memory Pipeline (Kenotic Labs), the first system designed against the ATANT framework, across 5 test suite iterations. All evaluations are LLM-independent.

6.1 Historical Progression

Table 5: Reference Implementation Progression

Suite	Date	Stories	Questions	CP8	Key Change
Legacy	Jan 2026	50		58%	Legacy scoring (with LLM)
Legacy+	Feb 2026	50		72%	Tuning gains
Legacy++	Feb 2026	50		58%	Regression from over-tuning
1.0	Feb 25	50	304/304	100%	New architecture (no LLM)
1.1	Feb 28	50	304/304	100%	All 10 CPs at 100%
1.2	Feb 28	100	656/671	98%	Stress test: 12 failures
2.0	Mar 1	100	671/671	100%	Structural matcher fixes
2.1	Mar 1	250	1,835/1,835	100%	Full scale isolated

The legacy pipeline hit a ceiling at 58% and suffered from whack-a-mole regressions: optimizing for one narrative pattern broke retrieval for another. The current architecture broke through this ceiling in days, demonstrating that continuity is an architecture problem, not a tuning problem.

6.2 Cumulative Results

Cumulative mode is where continuity is actually tested. When multiple life narratives coexist in the same database, the system must retrieve the correct fact for the correct context without cross-contamination.

Table 6: Cumulative Mode Results

Mode	Stories	Questions	CP8 Rate
Isolated (250)	250/250	1,835/1,835	**100.0%**
Cumulative (50)	50/50	304/304	**100.0%**
Cumulative (250)	~210/250	1,761/1,835	**96.0%**

6.3 Compliance Achieved

Level	Status	Tier
ATANT-Core	Pass	Gold
ATANT-Stress	Pass	Gold
ATANT-Cumulative	Pass	Gold
ATANT-Scale	In progress	Silver (96%)

6.4 Failure Analysis

Suite 1.2 failures (12 stories, 15 questions): The structural matcher failed on niche predicates questions about specialized hobbies (bonsai, falconry, cave exploration) where the predicate vocabulary was too specialized. The fix (Predicate Lexicon expansion) was architectural, not parameter tuning.

250-story cumulative failures (74 questions): Similarly-named predicates from different stories compete when 250 narratives coexist. The system must disambiguate by context, entity, and trace convergence. The 4% gap represents the current frontier.

CP4 failures (Type Tagging, 51.6%): Object type tagging fails on exotic domain-specific objects ("varroa mite," "Paraloid B-72 adhesive"). These are diagnostic they do not affect CP8 accuracy.

7 Discussion

7.1 Continuity Is an Architecture Problem

The legacy-to-current progression (58% → 100%) demonstrates that continuity cannot be achieved through scoring optimization alone. The legacy pipeline suffered from regressions under tuning pressure a signature failure of systems without architectural continuity support. The breakthrough came from grammar-first classification, deterministic trace convergence, and structural matching architectural decisions, not hyperparameter changes.

7.2 Cumulative Mode Is the Real Test

Isolated mode proves the pipeline works. Any reasonably engineered system should eventually pass. Cumulative mode tests what retrieval-based systems fundamentally struggle with: disambiguation under memory load. When 250 life narratives share storage, semantic similarity conflates similar-but-distinct events. This is the gap between retrieval and continuity.

7.3 Model Agnosticism as Future-Proofing

ATANT evaluates continuity without any model in the loop. This is not a limitation it is a design principle. The intelligence layer is not static: today it is LLMs, tomorrow it may be vision models, world models, or embodied agents. A standard tied to any specific architecture dies when that architecture is superseded. The 7 properties and 10 checkpoints describe what continuity is, not what today's AI happens to look like.

7.4 Limitations

Keyword verification, not reconstruction quality. CP8 checks whether expected keywords appear in the retrieved answer. A system could pass by returning relevant facts without coherence. Future versions should add reconstruction quality metrics.

Single-author corpus. All 250 stories were written by one author, limiting linguistic diversity and cultural representation.

Single evaluated system. Only one system has been evaluated against ATANT. The framework's value depends on independent systems being tested. We invite any team building AI continuity to run ATANT and publish results.

English only. The corpus does not test multilingual continuity.

8 Conclusion

We presented ATANT, an evaluation framework for AI continuity the missing layer between AI interaction and AI relationship. ATANT provides a formal definition of continuity (7 properties), a model-independent evaluation methodology (10 checkpoints), a narrative test corpus (250 stories, 1,835 questions), and a sequenced compliance framework (4 levels). The reference implementation demonstrates that continuity is an architecture problem solvable through deterministic engineering, not probabilistic tuning.

ATANT is not frozen. Version 1.0 defines the foundation. Future versions will add reconstruction quality metrics, multi-language narratives, proactive behavior testing, and community-contributed stories. The standard grows as the field grows.

The framework, corpus, and evaluation protocol are available at https://github.com/Kenotic-Labs/ATANT.

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